

VMS System

Request -> Validate -> Approval

15 Governance Scenarios
Dataflows & Sample Workflows

Every action runs on a double-validation, single-approval spine, with separation of duties:

REQUEST -> VALIDATE (1) -> (work) -> VALIDATE (2) -> APPROVAL

requester != validator != approver

For Capstone Project Defense

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A. Core Repair Pipeline

1 Standard reactive repair (the full happy path)

```
REQ:      [Sarah/Custodian files report] -> [Paul/Admin creates ticket]
VALIDATE(1): [Sarah inspects -> "Needs Maintenance" -> sub-issue] -> [Paul assigns James]
           [James repairs]
VALIDATE(2): [Sarah functional test -> all Pass -> Approved]
APPROVAL:  [Paul confirms -> Done] -> [Paul closes ticket] -> vehicle Available
```

SAMPLE WORKFLOW

Sarah notices Ambulance 04's brakes feel soft and files an Issue Report (severity High). Paul creates Ticket #8 and assigns Sarah to inspect. She confirms it's real ("Needs Maintenance"), logs sub-issue "Brake pads." Paul assigns James, who replaces the pads and logs the work. Sarah road-tests it -- all checks pass, she Approves. Paul gives final Confirmation; Ticket #8 closes and Ambulance 04 returns to Available.

2 False alarm (Validate (1) kills the request)

```
REQ:      [Lisa files report] -> [Paul creates ticket]
VALIDATE(1): [Lisa inspects -> "No Issues" -> NO sub-issue created]
APPROVAL:  [Paul confirms & closes] -> Issue Report -> Resolved
```

SAMPLE WORKFLOW

Lisa reports "Firetruck 2 overheating." Paul opens Ticket #9. On inspection, Lisa finds the temperature gauge normal and coolant full -- a sensor glitch, not a real fault. She submits "No Issues" (no sub-issue). Paul reviews, agrees, and closes the ticket with no repair. The report is marked Resolved.

3 Multiple sub-issues from one inspection (fan-out)

```
REQ:      [Sarah files "brake problem"] -> [Paul creates ticket]
VALIDATE(1): [Sarah inspects -> finds 2 root causes -> 2 sub-issues]
           [James -> pads] + [David -> fluid line]  <- two independent tracks
VALIDATE(2): [Sarah verifies each separately]
APPROVAL:  [Paul confirms each] - ticket closes only when BOTH Done
```

SAMPLE WORKFLOW

Sarah's brake inspection uncovers two causes -- worn pads and a leaking fluid line. She logs them as two sub-issues under Ticket #8. Paul assigns James to the pads and David to the line. Each is repaired, verified, and confirmed on its own timeline; the ticket stays Active until both hit Done, then Paul closes it.

B. Work Outside the Normal Pipeline

4 Emergency field repair (work-first -- engine dies mid-drive)

```
WORK:      [Driver calls James] -> [James repairs in field, logs Standalone Record]
           progress = Assigned (NOT Completed) - personnel forced to James himself
VALIDATE:  [Precious inspects the repair -> Passed]
APPROVAL:  [Paul confirms -> progress = Completed] -> vehicle Available
```

SAMPLE WORKFLOW

Ambulance 06's engine fails on a route; the driver phones James, who does a field repair. James logs a Standalone Maintenance Record -- but the system parks it at "Assigned," not done. Precious (a different person) inspects and verifies it Passed; only then can Paul confirm it Completed. James cannot certify his own work -- the Request->Validate->Approval rule holds even when the work jumps the queue.

5 Preventive scheduled maintenance (the one no-gate exception)

```
REQ:  [Paul schedules "Oil change, every 6 mo"]
WORK: [James marks Done] -> Record = Completed IMMEDIATELY + next occurrence auto-created
```

SAMPLE WORKFLOW

Paul schedules a recurring 6-month oil change for Ambulance 04, assigned to James. On the due date James performs it and clicks "Mark Done" -- the record is instantly Completed (no verify/confirm), and the system auto-queues the next one for 6 months out. Deliberately lightweight: routine, low-risk, planned work needs no independent validation.

C. Rework & Correction Loops

6 Rejected repair (Validate (2) bounces it back)

```
[James repairs] -> VALIDATE(2): [Sarah functional test -> a check FAILS -> Rejected]
-> sub-issue -> Under Repair (back to James) -> [James redoes it] -> re-verify
```

SAMPLE WORKFLOW

James replaces the pads, but on Sarah's road test the brakes still pull left -- she marks that check Fail, so the whole verification is Rejected. The sub-issue drops back to Under Repair automatically and James is notified to redo it. No approval is ever reached until a clean pass.

7 Add sub-issue mid-repair (scope grows during work)

```
[James, repairing brakes, spots a leaking differential]
REQ: [James adds a new sub-issue on the SAME ticket] -> status Open
-> [Paul assigns it out] -> its own validate->approve track
```

SAMPLE WORKFLOW

While under Ambulance 04 for the brakes, James notices the rear differential leaking oil -- unrelated. He can't open a new ticket (Admin-only), so he adds a sub-issue "Differential leak" onto Ticket #8. Ticket progress recalculates (now 1 of 2); Paul assigns it and it runs its own cycle. Only the ticket's own Custodian/Mechanic can add -- never Admin.

8 Reassign mechanic mid-repair

```
[James is 60% done, falls ill] -> [Paul reassigns to David]
work log preserved - [David] continues -> verify -> approve
```

SAMPLE WORKFLOW

James is halfway through a transmission rebuild on Firetruck 2 when he's out sick for two weeks. Paul reassigns the sub-issue to David, who reads James's work log, finishes the job, and submits for verification. The audit trail shows both mechanics worked on it.

9 Admin unconfirms a Done sub-issue (post-approval adjustment)

```
[Sub-issue is Done/Confirmed] -> [Paul realizes cost was wrong]
-> [Paul clicks "Unconfirm" -> status back to For Confirmation]
-> [Paul re-confirms after fixing] <- stays in Admin's hands, no re-verify
```

SAMPLE WORKFLOW

After confirming the brake repair, Paul spots he entered the wrong maintenance cost. He clicks "Unconfirm" on the Done sub-issue, which pops it back to For Confirmation, corrects the figure, and re-confirms -- without dragging Sarah through another verification.

D. Decisions & Closures

10 Defer -- can't fix now (Approval of a decision)

```
VALIDATE(1): [Sarah inspects -> real problem]
APPROVAL: [Paul defers with reason] -> sub-issue -> Deferred
-> auto-opens a follow-up Issue Report (breadcrumb)
```

SAMPLE WORKFLOW

Sarah confirms Firetruck 2 needs a new water pump, but it's on back-order for a month. Paul records a Deferral ("part unavailable, ETA 4 weeks"). The sub-issue is marked Deferred and the system auto-creates a follow-up report so it resurfaces later.

11 Decision close (close early, defer the leftovers)

```
[Ticket has 3 sub-issues, only 1 Done]
APPROVAL: [Paul "Close as Decision"] -> ticket Closed
- the 2 unfinished sub-issues auto-marked Deferred
```

SAMPLE WORKFLOW

Ambulance 04's ticket has 3 sub-issues; only the brakes are done, and the barangay needs the vehicle back now. Paul closes the ticket as a Decision -- the brake fix stands, the other two are recorded as Deferred with his reason, and everyone (Sarah + mechanics) is notified.

12 Admin resolves an issue report directly (no ticket)

```
REQ:      [Lisa files a minor report]
APPROVAL: [Paul edits it straight to Resolved]  <- Admin authority, skips formal validate
```

SAMPLE WORKFLOW

Lisa reports a loose wiper that she then tightens herself. Paul, seeing it's trivial and already handled, edits the report straight to Resolved without ever opening a ticket. Only Admin can short-circuit like this; a Custodian may only touch their own still-Pending report.

E. Asset Lifecycle

13 Decommission (Validate = a guard rule)

```
REQ:      [Paul requests decommission + reason]
VALIDATE: [System: any open ticket on this vehicle?] -> if yes, BLOCKED
APPROVAL: [Paul] -> Decommissioned - removed from all coverage counts
```

SAMPLE WORKFLOW

Firetruck Uno's chassis is rusted beyond safe repair. Paul clicks Decommission, but the system blocks him -- there's still an Active ticket on it. He closes that as a Decision first, then decommissions with a reason. It drops out of the readiness/coverage math; history is kept.

14 Recommission

```
[Paul restores a Decommissioned vehicle] -> Available/Good - decommission fields cleared
```

SAMPLE WORKFLOW

A month later the barangay funds a full refurbishment of Firetruck Uno. Paul clicks Recommission; it returns to Available/Good and rejoins the active fleet.

15 Readiness check (pre-deployment validate -- the proactive gate)

```
REQ:      (vehicle about to be relied on for emergency response)
VALIDATE: [Precious runs the readiness checklist] -> any Fail = NOT Ready
APPROVAL: all Pass -> "Verified Ready to Respond" (valid 24 hrs)
```

SAMPLE WORKFLOW

Before the shift, Precious runs Ambulance 06's readiness checklist -- fuel full, oxygen present, siren works, stretcher aboard. All pass, so it's marked Verified Ready to Respond on the dashboard. This is the gate meant to catch a problem before a mid-drive failure like Scenario 4.